WORKED SOLUTION

STEP III 2013, Question 7

Pure • Energy Methods for ODEs

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 8.2, *Bounding Without Solving*: https://beyondstep.keithmcnulty.org/ch08/bounding-without-solving.html

The key idea

Each equation looks like a particle on a spring: a "force" term depending only on the unknown, plus (in (ii) and (iii)) a friction term proportional to the derivative. Multiplying the equation by the derivative turns it into the derivative of an "energy". With no friction the energy is conserved; with friction that never pushes the wrong way, the energy can only decrease, so it never exceeds its starting value. Part (iii) asks us to build the energy ourselves and to show that the friction coefficient is never negative.

Solution

Part (i)

Write $y' = \frac{\mathrm{d}y}{\mathrm{d}x}$. By the chain rule,

$$\begin{aligned}\frac{\mathrm{dE}}{\mathrm{d}x} &= 2y'y'' + 2y^3y' \\ &= 2y'\left(y'' + y^3\right) = 0,\end{aligned}$$

using the differential equation. So E is constant, and its value is $\mathrm{E}(0) = 0^2 + \frac{1}{2}\cdot 1^4 = \frac{1}{2}$. Hence for every x,

$$\tfrac{1}{2}y^4 = \tfrac{1}{2} - (y')^2 \le \tfrac{1}{2},$$

since $(y')^2 \ge 0$. So $y^4 \le 1$, which gives $|y(x)| \le 1$ for all x.

Part (ii)

Differentiating in the same way, and then using the equation in the form $v'' + \sinh v = -xv'$,

$$\begin{aligned}\frac{\mathrm{dE}}{\mathrm{d}x} &= 2v'v'' + 2\sinh v\, v' \\ &= 2v'\big(v'' + \sinh v\big) \\ &= -2x\,(v')^2.\end{aligned}$$

For $x \geq 0$ both x and $(v')^2$ are non-negative, so $\frac{\mathrm{dE}}{\mathrm{d}x} \leq 0$, as required.

So E is non-increasing on $x \geq 0$ and $\mathrm{E}(x) \leq \mathrm{E}(0)$ there. At $x = 0$, $\cosh(\ln 3) = \frac{1}{2}\big(3 + \frac{1}{3}\big) = \frac{5}{3}$, so $\mathrm{E}(0) = 0 + 2 \cdot \frac{5}{3} = \frac{10}{3}$. Therefore, for $x \geq 0$,

$$2\cosh v(x) = \mathrm{E}(x) - (v')^2 \leq \mathrm{E}(x) \leq \tfrac{10}{3},$$

so $\cosh v(x) \leq \frac{5}{3}$.

Part (iii)

We look for an energy of the form $\mathrm{E} = (w')^2 + F(w)$, as in (ii). Then

$$\frac{\mathrm{dE}}{\mathrm{d}x} = 2w'w'' + F'(w)\,w' = w'\big(2w'' + F'(w)\big),$$

and we want the bracket to become a multiple of w' via the equation. That happens if $F'(w) = 2(w\cosh w + 2\sinh w)$.

? **Ask yourself** What function of w has derivative $w\cosh w + 2\sinh w$?

Integrating by parts, $\int w\cosh w\,\mathrm{d}w = w\sinh w - \cosh w$, and $\int 2\sinh w\,\mathrm{d}w = 2\cosh w$. Adding,

$$\frac{\mathrm{d}}{\mathrm{d}w}\big(w\sinh w + \cosh w\big) = \sinh w + w\cosh w + \sinh w = w\cosh w + 2\sinh w.$$

So we take

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}w}{\mathrm{d}x}\right)^2 + 2\big(w\sinh w + \cosh w\big).$$

Writing $g(x) = 5\cosh x - 4\sinh x - 3$, the equation says $w'' + (w\cosh w + 2\sinh w) = -g(x)\,w'$, so

$$\frac{\mathrm{dE}}{\mathrm{d}x} = 2w'\big(w'' + w\cosh w + 2\sinh w\big) = -2g(x)\,(w')^2.$$

? **Ask yourself** Is $g(x)$ ever negative?

In exponentials,

$$\begin{aligned} g(x) &= \tfrac{5}{2}\left(\mathrm{e}^{x}+\mathrm{e}^{-x}\right)-2\left(\mathrm{e}^{x}-\mathrm{e}^{-x}\right)-3 \\ &= \tfrac{1}{2}\mathrm{e}^{x}-3+\tfrac{9}{2}\mathrm{e}^{-x} \\ &= \tfrac{1}{2}\mathrm{e}^{-x}\left(\mathrm{e}^{2x}-6\mathrm{e}^{x}+9\right)=\tfrac{1}{2}\mathrm{e}^{-x}\left(\mathrm{e}^{x}-3\right)^{2} \;\geq\; 0 \end{aligned}$$

for every x (it is zero only at $x=\ln 3$). Hence $\frac{\mathrm{dE}}{\mathrm{d}x}\leq 0$, and $\mathrm{E}(x)\leq \mathrm{E}(0)$ for $x\geq 0$. At $x=0$, $w=0$ and $w'=\frac{1}{\sqrt{2}}$, so $\mathrm{E}(0)=\frac{1}{2}+2(0+1)=\frac{5}{2}$. Therefore, for $x\geq 0$,

$$2\left(w\sinh w+\cosh w\right)\leq \mathrm{E}(x)\leq \tfrac{5}{2}, \qquad \text{so} \qquad w\sinh w+\cosh w\leq \tfrac{5}{4}.$$

Finally, $w\sinh w\geq 0$ for every real w, because w and $\sinh w$ always have the same sign. Dropping this term gives $\cosh w(x)\leq \frac{5}{4}$ for $x\geq 0$, as required.

Checks

- In (ii) the bound is sharp at the start: $\cosh v(0)=\cosh(\ln 3)=\frac{5}{3}$.
- In (iii), $g(0)=5-0-3=2$, which agrees with $\frac{1}{2}(1-3)^2=2$ from the factorised form.
- A numerical solution of each equation confirms the picture: in (i) E stays at 0.5; in (ii) E falls from $\frac{10}{3}$; in (iii) E falls from 2.5 and $\cosh w$ never rises above about 1.03, well inside the bound 1.25.

Common slips

- Forgetting the factor w' from the chain rule when differentiating $F(w)$ with respect to x.
- In (iii), stopping at $w\sinh w+\cosh w\leq\frac{5}{4}$ without explaining why $w\sinh w\geq 0$ lets us drop that term.
- Claiming $g(x)>0$: it vanishes at $x=\ln 3$, so only $\leq$ holds (which is all we need).

Beyond STEP

These energy functions are *Lyapunov functions*: quantities that never increase along solutions, used to prove that a system stays bounded or settles down without solving it. The same idea, conservation of energy plus dissipation by damping, is the basis of stability theory for nonlinear oscillators. Boyce and DiPrima, *Elementary Differential Equations*, covers Lyapunov's method in its chapter on nonlinear systems and stability.

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